Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



## Team Details

Project: LingoLive — Understand. Reply. Keep going.

Team name: CreativeEngine

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Problem: Help immigrants and interstate newcomers respond in an unfamiliar language.

Theme: Media, Content & Digital Experiences | Category: Education

## Brief about the idea

# Find your words in a new city.

A newcomer in Bengaluru needs to answer a cab driver in Kannada. LingoLive helps them understand, choose a reply and say it aloud.

### Understand

Live multilingual speech → meaning

### Reply

Useful options + pronunciation

### Learn later

Saved words and personal practice

# Opportunities

## How is it different?

Connects the next turn, a pre-task plan, relevant personal/place memory, and later learning in one small interface.

## How does it solve the problem?

Prioritizes usable replies during the encounter; the learner reviews word meanings afterwards. Repeated replies do not become new questions.

## USP and intended users

Local, editable context follows the learner. Initial pilots: relocation teams, universities and community language groups. Outcomes and demand are not yet measured.

# List of features offered by the solution

### Hear and respond
Google multilingual transcription; three short replies.

### Say it your way
Express familiar-language intent in the learning language.

### Recognize readbacks
Fuzzy repeat guards + session voice estimates; correction.

### Plan before leaving
Cab → optional tea → station; explicit progress controls.

### Relevant memory
Editable routines; opt-in saved-place notes on return.

### Learn afterwards
Saved review, word bubbles and pronunciation.

### Minimal interface
One large microphone control; typing and first-use tutorial.

### Bounded access
Private host code or your own Google key; usage guards.

# Process flow diagram or Use-case diagram

### 1. Listen

16 kHz browser microphone

→

### 2. Transcribe

Gemini Transcribe Live

→

### 3. Route the turn

Text/voice repeat estimates

### Your repeated words

Keep in local conversation context. Wait for the other speaker. Uncertain meaning → confirm the speaker.

### Other / unknown turn

Two Flash-Lite branches → meaning + replies → learner speaks → Saved learning later.

Speaker estimates help routing; they are not reliable identity. Explicit correction always wins.

# Wireframes/Mock diagrams of the proposed solution (optional)

## Conversation

Heard words

Meaning in your language

Three usable replies

Original / Pronunciation

Large listen button

## Before the task

Where do you want to go?

Optional stops

Learner-controlled progress

Relevant local memories

## Saved learning

Your conversation library

Tap a word → meaning

Pronunciation and practice

No new lesson request

Functional prototype screens are shown in the snapshots section and video.

# Architecture diagram of the proposed solution

## Browser

- AudioWorklet capture
- Local repeat / voice guard
- Conversation + plan context
- IndexedDB saved lessons
- Local, opt-in place memory

⇄

## Google Cloud Run

- FastAPI HTTP + WebSocket
- Host code / personal-key session
- Request + audio quotas
- Exact-source validation
- Secret Manager access code

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## Google AI

- Gemini 3.5 Transcribe Live
- Gemini 3.1 Flash-Lite
- Source and reply branches
- Vertex service account
- BYOK: Gemini Developer API

Firebase / Firestore are not used in this prototype. Any future cloud sync would require a separate opt-in design.

# Technologies to be used in the solution

## Google AI

Gemini 3.5 Transcribe Live; two parallel Gemini 3.1 Flash-Lite branches with minimal thinking.

## Google Cloud

Cloud Run, Vertex AI, Secret Manager, Cloud Build and Artifact Registry.

## Web and backend

Responsive JavaScript PWA, AudioWorklet, IndexedDB, Python FastAPI and WebSockets.

## Local adaptation

Bounded context and lexical/acoustic heuristics; no model-weight fine-tuning or stored voiceprint.

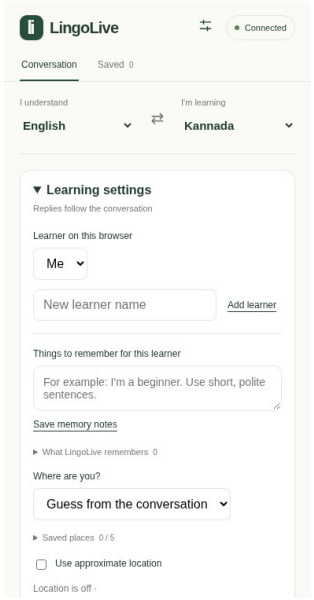
## Estimated implementation cost (optional)

# Cost model: pay for useful turns, bound prototype usage.

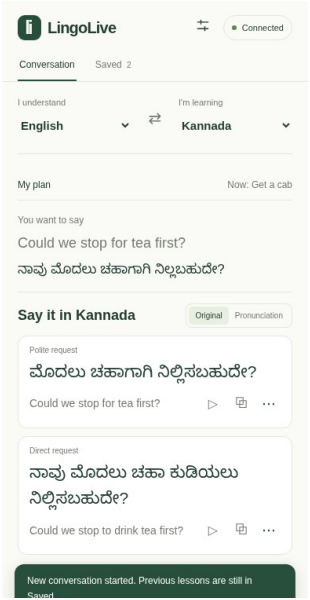
- A 15-minute listening session uses up to 15 minutes of streaming audio.
- N new lessons use 2 × N Flash-Lite generation calls.
- Detected repeats, saved word review and script switching add no lesson call.
- Cloud Run: 1 CPU / 512 MiB, request billing, min 0 / max 1 instance.
- Host limits: 400 raw model calls/day and 60 audio minutes/day.
- Prices depend on tokens, provider rates and actual runtime; no dollar estimate is asserted.

Quotas are prototype safeguards, not a hard billing cap. A wider launch needs durable per-user budgets.

# Snapshots of the prototype




Plan before leaving




Find a useful reply




Learn afterwards

Recorded prototype views. Word-review screenshot uses labelled test data; the video includes real Google responses.

# Prototype Performance report/Benchmarking

## Functional evidence, with clear measurement limits.

| | |
|---|---|
| **Earlier v1.5 warm HTTP lessons** | 10 complete lessons · mean 1.79 s · historical sample |
| **Later v1.9 typed HTTP checks** | 4 functional cases · mean 2.065 s · not a latency guarantee |
| **Current readback guard** | 25 local checks + 10 browser-flow checks; repeats add no lesson |
| **Voice estimates** | 3/4 held-out clean synthetic learner clips recognized; 0/4 others labelled user |
| **Service checks** | 112 backend tests; real Google replies and current Cloud Run guards verified |

No measured learning-outcome, noisy-phone, cold-start or population speaker-accuracy claim.

# Additional Details/Future Development (if any)

## Social value and business case

Help newcomers participate in everyday encounters.

Pilot with relocation teams, universities and community language groups.

Proposed organization-sponsored access; validate demand before expansion.

## Next engineering work

Representative mobile/noisy-language studies.

Durable per-user budgets and stronger identity.

Opt-in cross-device memory, potentially Firestore.

Measure task completion, reply usefulness and learning retention.

Current limits: ASR can mishear; voice estimates can be wrong; local profiles are not private cloud accounts.

# Provide links to your:

GitHub source — github.com/U4AR/LingoLive

Demo video — 3–4 minute working demonstration

Final product — live Google Cloud Run prototype

Submission materials — pitch, technical PDF and entry text

Keep the source repository publicly accessible throughout evaluation.

AI access: use your Google key, or obtain the host code privately from the team. No keys or codes are included.

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# Thank you

